

Aether One™ – ASIC Implementation Plan 2026

Engineering Sample Preparation for TrustCore NX (7 nm FinFET)

Objective

Transition the TrustCore NX prototype into a silicon implementation. The 2026 Engineering Sample will realize a 32 TOPS RVV-accelerated SoC with integrated AI and PQC engines, based on the validated software prototype (Aether OS, TVM, OpenSSL PQC Provider).

1. RTL Finalization (Q4 2025 – Q1 2026)

- Implement RVV 1.0a ISA support and FP16 (Zfh) extensions.
- Integrate Neural Coprocessor with TVM MetaSchedule-derived RVV kernels (terra_fusion_rvv_optimized.tar).
- Embed Kyber and Dilithium accelerators from liboqs_rvv for PQC operations.
- Add AMBA AXI4/ACE bus interfaces for Aether OS memory access.
- Design adaptive clock domains (0.8 – 1.1 V) for AI and crypto loads.

2. RTL → Gate-Level Implementation (Synopsys / Cadence)

Tools: Synopsys Design Compiler, ICC II, PrimeTime; Cadence Innovus / Tempus.

Inputs: RVV and PQC cores (SystemVerilog), TVM/PQC test vectors (JSON, Python generated).

Outputs: 7 nm netlist and layout (~150 mm²), timing and power reports, DRC/LVS verification results.

3. Validation & Verification (Engineering Sample Stage)

- Logic Simulation: apply TVM/PQC test vectors to RTL via Python → SystemVerilog interface.
- Gate-Level Simulation: verify RVV instruction execution (VLEN, mask ops).
- FPGA Prototype: Xilinx Versal or Intel Agilex board for firmware bring-up.
- Power/Thermal Analysis: simulate 32 TOPS load, 6 W TDP target.

4. Engineering Sample (Tape-Out Target – Q3 2026)

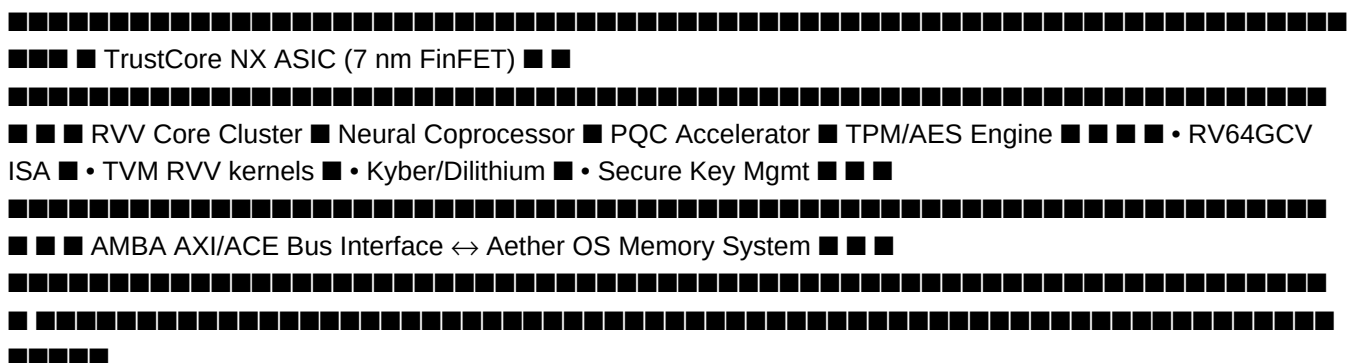
Foundry: TSMC 7 nm FinFET (ES lot ≈ 25 dies). Objectives: full RVV 1.0 compatibility, on-chip Kyber/Dilithium acceleration, integrated TPM 2.0/AES-512 engine, AI + PQC co-execution under 6 W power.

5. Validation Data Reuse

Existing prototype artifacts used as RTL verification inputs:

- terra_fusion_rvv_optimized.tar → RVV vector kernels for simulation.
- aether_pqc_benchmark.json → latency and throughput metrics for PQC hardware validation.

6. ASIC Block Overview



7. Summary & Next Steps

- Finalize RVV/PQC IP interfaces (AXI & Clock Domain Crossing).
- Generate RTL Verification Plan v1.0 (Python ↔ SystemVerilog adapter).
- Begin layout for Engineering Sample mask set (Q3 2026 target).
- Integrate Aether OS firmware with FPGA prototype for early bring-up.

This plan bridges the validated software prototype to hardware silicon, delivering a fully integrated

RVV■Accelerated Post■Quantum Secure AI platform.